TourSights - Tourism Intelligence

Digital Egypt Youth Initiative (مبادرة رواد مصر الرقمية)

CLS Learning Solutions

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and Requirements Gathering.
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Executive Summary

Problem. Tourists publish large volumes of opinions about Egyptian museums and archaeological sites on online review platforms. These opinions are unstructured, may be written in several languages, and mix praise and complaints about different parts of a visit. Reading them manually across platforms and destinations makes it hard to see which problems recur, how destinations compare, and whether complaint patterns change over time.

Intended users. The principal intended user is a decision-maker at the Egyptian Ministry of Tourism and Antiquities. Destination managers and tourism analysts are secondary beneficiaries. Direct access to Ministry personnel has not been confirmed, so requirements are based on the project brief and intended use.

Proposed solution. TourSights is an academic proof-of-concept that converts public online reviews into a structured, comparable dataset and presents it in an interactive Power BI Desktop report. It is not a deployed national monitoring platform, a web application, or a commercial product. The system supports decision-making; it does not replace tourism expertise or prescribe operational interventions.

Destinations. The intended scope is about five Egyptian museums and archaeological sites. Candidates include the Giza Pyramids and Sphinx, the Grand Egyptian Museum, Karnak Temple Complex, the Valley of the Kings, and Abu Simbel Temples. These are candidates only. The team will assemble a pool of about 8-10 destinations, test data feasibility on a subset of about three, and confirm the final five according to data volume, data quality, relevance, and remaining time.

Data collection. Two Outscraper exports generated on 19 October 2026 are confirmed as pilot data: 500 Tripadvisor reviews for The Pyramids of Giza and 220 Google Maps reviews for Giza Necropolis. These are raw extraction counts, not yet cleaned usable-review counts. Apify is planned as an additional collection tool, particularly to test the Tripadvisor Reviews Actor; no Apify run has yet been completed. Additional destinations and sources remain planned, subject to practical feasibility and available credits.

Role of Power Query and Power BI. Power Query is the primary tool for table-based cleaning, standardization, and preparation. Power BI provides the data model, DAX measures, slicers, one overview page, one main page per final destination, and hidden drill-through pages where feasible.

Optional NLP enhancement. Sentiment classification and aspect categorization will use a pretrained model and transparent keyword rules, validated against a manually labeled sample. No model will be trained from scratch. If the NLP step proves unreliable, a simpler documented fallback keeps the core dashboard deliverable.

Expected analytical outputs. (1) Most frequently mentioned problems per destination; (2) share of negative sentiment per experience category; (3) changes in complaint patterns over time, where dates support it; (4) positive aspects of the visitor experience; (5) comparable, evidence-based prioritization of issues, using explicit denominators.

Current status (19 October 2026). Two Outscraper pilot exports are available: 500 Tripadvisor reviews for The Pyramids of Giza and 220 Google Maps reviews for Giza Necropolis. Data cleaning, expanded collection, categorization, modeling, validation, and dashboard development remain planned. No model accuracy, final sentiment distribution, dashboard performance result, or lecturer feedback has been established.

Primary constraints. Approximately six weeks of implementation (proposed finalization 30 November 2026, to be confirmed against the institutional deadline), a four-member team with limited NLP experience, free tools only, and the representativeness limits of user-generated reviews.

List of Abbreviations

Abbreviation	Meaning
ABSA	Aspect-Based Sentiment Analysis
BI	Business Intelligence
DAX	Data Analysis Expressions
KPI	Key Performance Indicator
NLP	Natural Language Processing

Abbreviation	Meaning
CSV	Comma-Separated Values
FR / NFR	Functional / Non-Functional Requirement
MCIT	Ministry of Communications and Information Technology (logo source only)
MoTA	Ministry of Tourism and Antiquities (primary stakeholder)

Status labels used throughout this report: *Confirmed* (already true), *Planned* (to be done), *Proposed target* (a goal, not a result), *To be evaluated* (not yet measured).

1. Project Planning & Management

1.1 Project Proposal and Overview

Background. Online reviews are a widely studied source of customer and visitor opinion (Pang & Lee, 2008; Liu, 2012). For tourism, they offer unprompted descriptions of what visitors experienced. Overall star ratings summarize satisfaction but do not say why visitors are satisfied or dissatisfied, which is why text analysis is of interest (Xiang et al., 2015).

Problem. See Section 1.2.

Proposed solution. A Power BI report built on a cleaned, structured review dataset, optionally enriched with sentiment and visitor-experience categories.

Primary stakeholder. Decision-makers at the Egyptian Ministry of Tourism and Antiquities. Note the distinction: the logo in the document header belongs to the Ministry of Communications and Information Technology because it appears in the initiative's document branding; it does not indicate that this ministry is the project's stakeholder.

Intended beneficiaries. Destination managers and tourism analysts (secondary).

Motivation and expected value. The expected value is faster, more consistent, and more transparent insight into recurring visitor issues and positive aspects. This value is expected, not demonstrated. The project has not improved tourist satisfaction or influenced any decision.

Overall approach. Select destinations by data feasibility; collect with Outscraper; clean with Power Query; optionally enrich with NLP; model and visualize in Power BI; test and document.

1.2 Problem Statement

Tourist opinions are spread across platforms, written in free text, possibly in several languages, and cover different aspects of a visit (price, queues, staff, facilities, and so on). Manual reading does not scale, and it makes it difficult to identify recurring complaints, compare aspects of visitor experience, recognize positive aspects, and monitor change over time. TourSights aims to consolidate available review data into an organized analytical dataset and present it through an interactive Power BI dashboard, using data cleaning, descriptive analytics, sentiment analysis, and aspect-based categorization, subject to technical feasibility and validation.

Many reviews are vague (for example, "the facilities were poor" without naming a facility). The project therefore does not attempt to diagnose a precise physical location within a site unless a review explicitly supports it.

1.3 Project Objectives

General objective. Build a Power BI prototype that turns public visitor reviews for selected Egyptian museums and archaeological sites into structured, comparable indicators that support evidence-based discussion of visitor experience.

Specific objectives (implementation).

ID	Objective	Measurable success criterion (proposed target)
O1	Define scope, requirements, and destination selection procedure	Documented and reviewed by all members by end of Week 1
O2	Collect reviews with Outscraper and evaluate Apify where useful	Usable count recorded per destination and platform; target about 500-1,000 where feasible

ID	Objective	Measurable success criterion (proposed target)
O3	Clean and standardize data in Power Query	Documented, repeatable transformation steps; field completeness reported
O4	Handle technical duplicates without removing independent reviews	Duplicate rule documented; counts of flagged records reported
O5	(Optional extension) Add sentiment and aspect categories	Validated against a manually labeled sample; metrics reported only after measurement
O6	Deliver a Power BI report with overview and per-destination pages	Pages, slicers, and measures present and tested
O7	Test and document the prototype	Test table completed with actual outcomes

Expected analytical value (not an implementation objective). Identification of frequent issues, negative rates by category, trends, positive aspects, and normalized comparisons. These are intended outputs; no values have been produced yet.

1.4 Scope and Boundaries

In scope.

- Approximately five Egyptian museums and archaeological sites, confirmed through the feasibility procedure.
- Outscraper-based initial collection: two raw pilot exports are available (500 Tripadvisor reviews for The Pyramids of Giza and 220 Google Maps reviews for Giza Necropolis). These are raw extraction counts; cleaning and deduplication remain pending.
- Power Query-based cleaning and standardization.
- Sentiment and aspect categorization as an analytical extension.
- An interactive Power BI Desktop dashboard (static analytical snapshot).
- Data quality checks, model evaluation, testing, and documentation.

Out of scope for the current version.

- Continuous production monitoring.
- Automatic scheduled data collection and refresh.
- A fully deployed public web application.
- Training a large language model from scratch.
- Comprehensive coverage of all Egyptian tourist destinations.
- Guaranteed access to all online review platforms.
- Automatic prescription of expert tourism interventions.
- Diagnosing the precise physical location of a problem when review text does not state it.

Destination selection procedure (planned).

1. Prepare a pool of about 8-10 candidate destinations.
2. Assess availability and relevance of review data.
3. Run feasibility tests on a subset of about three destinations.
4. Confirm the final five using data volume, data quality, relevance to visitor experience, and remaining time.

Candidate (not final)	Status
Giza Pyramids and the Sphinx	Pilot exports: 500 Tripadvisor reviews for The Pyramids of Giza and 220 Google Maps reviews for Giza Necropolis (both via Outscraper, 19 Oct 2026); raw counts not yet cleaned
Grand Egyptian Museum	Candidate; no data collected
Karnak Temple Complex	Candidate; data not yet collected
Valley of the Kings	Candidate; data not yet collected
Abu Simbel Temples	Candidate; data not yet collected
Additional pool (3-5 more)	To be identified in Week 1

1.5 Proposed Solution

TourSights consolidates reviews into a table with fields such as destination, source, review text, publication date, rating, and language (where available). Power Query prepares the data; an optional Python step adds sentiment and aspect labels; Power BI models the data and presents it.

The dashboard differs from a conventional rating-average display because it aims to show *which aspects* of the visit are mentioned and how often they are associated with negative sentiment, always with explicit denominators and sample sizes.

1.6 Project Methodology

The planned pipeline, in order:

Step	Stage	Tool	Output
1	Destination selection	Excel, team review	Shortlist and final destination decision
2	Source feasibility testing	Outscraper and Apify documentation; small tests	Source and field availability matrix
3	Review collection	Outscraper; Apify Actors where tested and useful	Raw CSV/Excel files
4	Raw-data storage	Folder structure, collection log	Untouched raw copy
5	Power Query cleaning	Power Query	Standardized, typed table
6	Data-quality validation	Power Query, Excel	Quality report
7	NLP enrichment (if implemented)	Python, pretrained model	Sentiment and aspect output file
8	Sentiment and aspect categorization	Python + keyword rules	Review-level and aspect-level records
9	Data transformation and modeling	Power BI	Star-schema model
10	Measures and dashboard	DAX, Power BI	Report pages
11	Manual review and testing	Team	Test results
12	Final documentation and presentation	Word, PowerPoint	Final package

Architecture (text form): Outscraper export (CSV/Excel) → raw folder → Power Query (cleaning) → cleaned table → [optional] Python/NLP enrichment → enriched CSV → Power BI model (FactReviews, FactReviewAspects, dimensions) → DAX measures → dashboard pages. Power Query performs table-based preparation; advanced language processing happens in a separate step because Power Query does not provide general semantic NLP.

Duplicate handling principle. Independent reviews with similar or identical complaints are retained, because their frequency is itself analytical evidence. Only demonstrable technical duplicates (the same review included twice through repeated extraction or file merging) are removed or flagged, using stable review identifiers or other defensible matching rules.

1.7 Project Plan and Timeline

The schedule below is a *proposed* plan for an approximately six-week implementation period starting 19 October 2026, with provisional finalization on 30 November 2026. It is not a confirmed institutional deadline. Tasks may overlap.

Week	Dates	Main activities	Deliverable
1	19-25 Oct 2026	Confirm problem, scope, initial destinations; test sources and fields; define data schema; confirm initial categories	Approved project plan; data-feasibility results
2	26 Oct-1 Nov 2026	Collect reviews for pilot destinations; expand where data allows; document sources and limits	Initial destination datasets; source inventory

Week	Dates	Main activities	Deliverable
3	2-8 Nov 2026	Clean and standardize in Power Query; validate dates, ratings, languages, identifiers; resolve technical duplicates; refine categories	Cleaned analytical dataset
4	9-15 Nov 2026	Evaluate pretrained models; implement feasible sentiment/aspect approach; label validation sample; evaluate	Initial model outputs and evaluation (if successful)
5	16-22 Nov 2026	Build Power BI model and DAX measures; overview and destination pages; drill-through where feasible	Working dashboard prototype
6	23-29 Nov 2026	Validate dashboard and results; fix major issues; finish report and presentation; prepare demo	Validated prototype and final documentation

Target finalization: 30 November 2026 (provisional). The team must confirm the formal submission date. Learning and troubleshooting time is included in Weeks 2–5; collection, Power Query work, and dashboard development may overlap.

1.8 Milestones and Deliverables

Milestone	Planned period	Main activities	Deliverable	Completion criteria
M1 Plan and scope approved	Week 1	Scope, objectives, candidate pool, source tests	Project plan; source matrix	Scope and exclusions documented; owners assigned
M2 Raw data collected	Weeks 1-2	Outscraper collection; Apify Actors for small tests; collection log	Raw files per destination	Actual counts recorded per destination
M3 Cleaned dataset	Weeks 2-3	Power Query cleaning; quality checks	Cleaned table; quality report	Types standardized; duplicate rule documented
M4 NLP outputs (if feasible)	Weeks 3-4	Model choice; categories; manual labeling	Enriched CSV; evaluation notes	Metrics computed from real labeled sample
M5 Dashboard prototype	Weeks 4-5	Model, DAX, pages	.pbix prototype	Overview and destination pages open and work
M6 Validated final package	Week 6	Testing; report; demo	Final report, .pbix, presentation	Test table completed with actual outcomes

1.9 Task Assignment and Team Responsibilities

The team has four members with similar technical starting points. The work is divided into four phases. Each member owns one primary phase after the team confirms the assignment. The phases are areas of ownership, not isolated projects.

Project Phase / Primary Responsibility	Student ID	Student Name
Phase 1 - Project Planning, Research, and Requirements	21163471	Ahmed Khaled Ahmed Sayed
Phase 2 - Data Collection and Data Preparation	21075562	Ganna Adel Mahmoud Ibrahim
Phase 3 - NLP, Sentiment Analysis, and Analytical Modeling	21125342	Malak Mohamed Mohamed Abdelmoneam
Phase 4 - Power BI Dashboard, Testing, and Integration	21168429	Mohamed Ashraf Hassan Abdulhamid

Coordination principles.

1. Each student owns one primary phase and is responsible for planning, executing, documenting, testing, and handing over its outputs.
2. Every member contributes to integration, peer review, manual validation, debugging, and final delivery.

3. The Phase 1 owner maintains scope, requirements, source documentation, schedule, and report consistency, but does not write other members' technical sections. Each phase owner writes the report content for their own work; the final editor consolidates and checks terminology, formatting, and agreement with the actual implementation.
4. Phase 4 begins modeling and dashboard preparation as soon as reliable cleaned data exists, without waiting for every NLP feature.
5. Each major output receives review from at least one other member.
6. The team holds a short weekly review meeting and keeps a shared task tracker (a spreadsheet is sufficient; no paid subscription is needed).

Phase 1 - Project Planning, Research, and Requirements

Objective. Establish the academic, analytical, and organizational foundation before collection and modeling expand.

Ordered steps. (1) Define the problem and intended user. (2) Confirm objectives and boundaries, including the 8-10 destination pool and exclusions. (3) Conduct the literature review using credible academic and official technical sources. (4) Assess data sources: document Outscraper as the tool used for the pilot, verify which other sources it supports, check available fields, run small feasibility tests, record limits. (5) Define data and analytical requirements: essential fields, preliminary taxonomy, sentiment labels, the duplicate distinction, key analytical questions. (6) Prepare stakeholders, user stories, use cases, and requirements without claiming Ministry approval. (7) Establish the plan, milestones, risk register, and KPIs. (8) Document findings for the other members.

Item	Detail
Tools	Web browser and Google Scholar (research); official Outscraper and Apify (small Actor tests) and platform documentation; Excel (matrices, requirements); Word; shared folder; Outscraper and Apify (small Actor tests) interface for small feasibility checks
Inputs	Project brief; project concept; two initial Giza-area pilot exports (500 Tripadvisor and 220 Google Maps records); source documentation; team structure; six-week schedule
Deliverables	Problem statement; objectives; scope; destination shortlist; source matrix; initial data dictionary; literature review; stakeholder analysis; requirements; user stories; use cases; timeline; risk register; KPIs; methodology draft
Acceptance criteria	Problem, user, objectives explicit; scope and exclusions documented; selection procedure defined; source approach documented with limits; required outputs listed; key claims have verified references; every task has an owner, timeline, and output; no invented results or stakeholder feedback
Risks and fallbacks	Literature taking too long: prioritize the strongest sources. Source unusable: update matrix and propose alternatives. Scope too broad: reduce destinations or advanced features, keeping the Power BI core
Handover to Phase 2	Candidate list; collection approach; preliminary field list; data dictionary; quality expectations; sample-size target; source and field findings; decisions to confirm during the pilot
Timing	Primary: Week 1. Light updates: Weeks 2-6

Phase 2 - Data Collection and Data Preparation

Objective. Acquire relevant reviews, consolidate them, and prepare a reliable dataset with Outscraper, CSV/Excel, and Power Query.

Ordered steps. (1) Inspect the two existing Giza-area pilot exports: 500 Tripadvisor reviews for The Pyramids of Giza and 220 Google Maps reviews for Giza Necropolis, both collected through Outscraper. (2) Standardize a common schema (Review ID, destination, platform, collection tool, review text, publication date, collection date if available, rating, language, and source metadata), documenting source-specific differences. (3) Collect additional reviews with Outscraper; test the Apify Tripadvisor Reviews Actor on a small sample before relying on it; use the Apify Google Maps Actor only if it adds practical value. Target approximately 500-1,000 usable reviews per destination where feasible and record actual counts, platforms, tools, and date coverage. (4) Organize raw data: retain an untouched copy, use meaningful file names, and keep a collection log. (5) Import

into Power Query and standardize column names, types, destination/platform/tool labels, dates, ratings, missing values, and text formatting; document steps. (6) Retain independent reviews with similar complaints; remove or flag only technical duplicates by review ID/URL or defensible matching rules. (7) Check field completeness, mapping, duplicates, date validity, rating ranges, empty text, language, relevance, and sample-size differences. (8) Prepare a cleaned dataset with documented exclusions and source traceability.

Item	Detail
Tools	Outscraper and Apify (small Actor tests) (collection); Excel (inspection); CSV/Excel (exchange); Power BI Desktop and Power Query (cleaning); optional Python/pandas for supplementary inspection
Inputs	Pilot dataset; Phase 1 destination and source assessment; preliminary data dictionary; field-availability findings
Deliverables	Raw files by destination and source; standardized schema; Power Query workflow; cleaned dataset; actual counts; quality report; duplicate summary; missing-value summary; final data dictionary; source and date-coverage record; files ready for NLP; draft report text on acquisition, cleaning, and quality
Acceptance criteria	Files import; names and types standardized; destination and source identifiers present; duplicate handling documented; no review removed solely for similar wording; exclusions documented; actual sample sizes known; data usable downstream; quality checks run and recorded
Risks and fallbacks	Too few reviews: assess a replacement or reduce scope openly. Source unavailable: proceed with collected sources. Missing dates: limit trend claims. Unreliable metadata: fix mapping before comparisons. Time overrun: fewer destinations with better quality
Handover to Phase 3	Cleaned dataset; data dictionary; identifiers; language and date information; quality limitations; actual counts per destination; taxonomy and required output fields; notes on ambiguous or irrelevant reviews; stable record identifier where the source provides one
Timing	Primary: Weeks 1-3 (may overlap with early NLP preparation)

Phase 3 - NLP, Sentiment Analysis, and Analytical Modeling

Objective. Turn cleaned text into interpretable sentiment labels and visitor-experience categories, validate quality, and produce fields for Power BI, prioritizing feasibility and transparency over algorithmic complexity.

Ordered steps. (1) Inspect representative reviews per destination: languages, topics, positive/negative/neutral/mixed examples, ambiguous or sarcastic wording. (2) Finalize a documented taxonomy (proposed: Price and Value for Money; Cleanliness; Queues and Crowding; Accessibility and Transportation; Staff and Tour Guides; Vendor Behavior, Harassment, and Scams; Facilities and Visitor Services; Signage and Visitor Information; Safety and Security; Historical and Cultural Experience; Museum Exhibitions and Organization; Accessibility Needs where evidence allows), each with a definition, examples, and distinguishing guidance. (3) Select a feasible approach: evaluate a pretrained sentiment model for license, supported languages, compute needs, and fit for short reviews; do not assume benchmark scores transfer to this dataset. (4) Implement sentiment labels (Positive, Negative, Neutral, Mixed if supported), separating overall sentiment from aspect sentiment. (5) Implement aspect categorization using predefined categories, transparent keyword or phrase rules, the pretrained model, and manual review of ambiguous cases; keyword matching is not described as a trained aspect-based model. (6) Create a manually labeled validation sample with written annotation rules, independent labeling where practical, documented disagreement resolution, and a separate evaluation subset where feasible. (7) Evaluate against the labels using precision, recall, F1, macro F1, and a confusion matrix as applicable; analyze errors such as negation, sarcasm, mixed sentiment, language variation, unrelated comments, and wrong aspects. (8) Generate analytical fields with explicit denominators, distinguishing review counts from aspect-mention counts. (9) Export an output dataset joinable to the cleaned data by review identifier, leaving unsupported or uncertain cases flagged rather than filled.

Item	Detail

Item	Detail
Tools	Python; a pretrained sentiment model; Hugging Face Transformers if applicable (Wolf et al., 2020); Google Colab or similar; scikit-learn (Pedregosa et al., 2011) for evaluation; Excel for annotation; CSV transfer to Power BI
Inputs	Phase 2 cleaned dataset; review texts; language metadata; preliminary categories; objectives
Deliverables	Documented taxonomy; sentiment workflow; aspect categories per review where supported; aspect-level sentiment where supported; labeled validation subset; evaluation report with measured metrics only; error log; enriched dataset; indicator definitions; draft report text on method, validation, and limitations
Acceptance criteria	Approach implemented or fallback adopted and documented; taxonomy defined; identifiers allow matching; validation sample prepared; any metric is computed from real results; errors documented; output usable by Power BI; no unmeasured performance presented as established
Risks and fallbacks	Poor model: test a simpler baseline and report honestly. Weak multilingual results: restrict advanced classification to validated languages and report others as unsupported. Aspect sentiment too complex: combine aspect detection with review-level sentiment and state the limitation. Slow execution: smaller model or manageable sample. Frequent annotator disagreement: clarify rules and recheck
Handover to Phase 4	Cleaned and enriched datasets; category definitions; evaluation report with actual metrics; list of uncertain or unsupported classifications; metric definitions; usable Power BI fields; visualization guidance
Timing	Primary: Weeks 3-4; evaluation and refinement may continue in Week 5

Phase 4 - Power BI Dashboard, Testing, and Integration

Objective. Integrate prepared data into an interactive, professional Power BI report, validate calculations and usability, and finish documentation and presentation. The NLP component supports the dashboard but must not prevent its delivery.

Ordered steps. (1) Confirm the dataset and which measures can actually be calculated. (2) Import into Power BI and complete any structured Power Query transformations; confirm types and relationships. (3) Design the data model (FactReviews, FactReviewAspects, DimDestination, DimSource, DimDate, DimAspect, DimLanguage, adapted to real fields), ensuring one review linked to several aspects does not inflate review counts. (4) Define and validate DAX measures: usable reviews, counts per destination, sentiment distribution, aspect mentions, negative rate by aspect, positive aspects, rating statistics, trends where data allows, normalized comparisons, and documented priority indicators. (5) Build the overview page. (6) Build one main visible page per finalized destination. (7) Add slicers, page navigation, tooltips, and hidden drill-through pages where supported; hidden pages are a navigation choice, not data protection. (8) Test DAX and visuals: reconcile totals, verify denominators, test filters and drill-through, check for double counting and empty selections, record results honestly. (9) Validate interpretation: no destination labeled worse purely on raw counts; no trends from inadequate data; model confidence not treated as calibrated probability; priorities described as analytical indicators only. (10) Finalize documentation and demonstration.

Item	Detail
Tools	Power BI Desktop; Power Query; DAX; CSV/Excel; Word; PowerPoint; Git or another version-control method if useful
Inputs	Phase 2 dataset; Phase 3 enriched data if implemented; requirements; data dictionary and model definitions; destination set; actual validation findings
Deliverables	.pbix file; overview page; one page per finalized destination; hidden drill-through pages where implemented; documented Power Query steps; validated DAX measures; test report; usage explanation; integrated report; presentation; limitations and future-work list

Item	Detail
Acceptance criteria	File opens and runs on the project computer; overview and destination pages exist; counts reconcile with source tables; percentages have defined denominators; filters and drill-through work where implemented; missing data handled transparently; only actual values shown; tests run and outcomes recorded; report matches implementation; files organized
Risks and fallbacks	NLP unavailable: deliver the core dashboard on descriptive fields with reduced scope. Inadequate destination data: document the gap and use destinations meeting feasibility criteria. Slow performance: simplify visuals or model. Short on time: prioritize correct measures, overview, destination pages, testing, and documentation over decoration
Final handover	Final .pbix; cleaned and enriched datasets; documentation; data dictionary; test results; source inventory; known limitations; presentation; future-work notes
Timing	Primary: Weeks 4-6; development may begin in Week 4 on the cleaned base dataset

Cross-Phase Dependencies

From	Required inputs	Outputs	Recipient	Handover checkpoint	Fallback if delayed
Phase 1	Brief; pilot sample	Scope, candidates, source assessment, schema, collection plan	Phase 2	End of Week 1	Phase 2 proceeds with the Giza pilot and a provisional schema
Phase 2	Phase 1 outputs	Cleaned data, field documentation, language coverage, quality report	Phase 3	End of Week 3	Phase 3 works on the Giza pilot subset
Phase 2	Cleaned base data	Early data for modeling	Phase 4	Mid-Week 3 to Week 4	Phase 4 builds the model on partial data
Phase 3	Cleaned data; taxonomy	Validated sentiment and aspect outputs, indicator definitions (if implemented)	Phase 4	End of Week 4 (refinement in Week 5)	Dashboard uses descriptive fields and simpler documented categories
Phase 4	All outputs	Integrated dashboard, test findings, correction requests	All members	Weekly from Week 4	Fix the highest-priority issues first

Integration starts early rather than being left to the final week.

Workload Balance

The four phases are intended to carry broadly comparable effort once technical complexity and learning needs are considered. Equal effort is not claimed. Phase 1 is research- and documentation-heavy; Phase 2 is hands-on data work; Phase 3 carries the highest technical uncertainty; Phase 4 combines modeling, DAX, testing, and presentation. Collaboration is expected in manual annotation, resolving inconsistent destination data, debugging transformations, verifying DAX measures, and final testing and report consistency.

1.10 Resource Allocation and Budget

The project prioritizes free tools and available free usage allowances. Collection through Outscraper and Apify is subject to the current credits, source-specific pricing, and limits displayed by each service. The team must not assume that high-volume extraction remains free after the available allowance is consumed; if limits are reached, it will prioritize a smaller high-quality dataset rather than depend on paid services.

Tool	Purpose	Notes
Outscraper	Google Maps and Tripadvisor review collection	Pilot exports: 500 Tripadvisor reviews and 220 Google Maps reviews for Giza-area listings (19 Oct 2026). Tripadvisor service lists first 500 reviews/month free, then usage pricing; verify current quota/pricing before more collection (Outscraper, n.d.-b).

Tool	Purpose	Notes
Apify	Additional collection tool; evaluate Tripadvisor Reviews Scraper and optionally Google Maps Actor	Free plan lists $5 monthly usage credit; Tripadvisor Actor lists from $0.90 per 1,000 reviews. Actual run, cost, and suitability not yet tested; prices may change (Apify, n.d.-b, n.d.-c).
Power BI Desktop	Model, DAX, dashboard	Desktop application; verify licensing for each member
Power Query	Cleaning and transformation	Built into Power BI and Excel
Python and open-source packages	Optional NLP	Verify package and model licenses
Google Colab or similar	Free notebook environment	Subject to current free-tier limits
Excel / CSV	Intermediate files, annotation	
Git/GitHub (optional)	Version control	Optional

Workstream	Phase owner	Support
Coordination, documentation, research	Phase 1	All (review)
Source assessment and collection	Phase 2 (with Phase 1 input)	Phase 1
Data preparation, Power Query, modeling inputs	Phase 2	Phase 4
NLP exploration, validation, analytical methods	Phase 3	All (annotation)
Dashboard, DAX, testing, presentation	Phase 4	All (testing)

1.11 Risk Assessment and Mitigation Plan

Probability and impact are qualitative judgments, not statistical estimates.

ID	Risk	Probability / Impact / Priority	Preventive measures	Contingency / fallback	Responsible phase
R1	Insufficient data for some destinations	Medium / High / High	Early feasibility tests; maintain a candidate pool of 8-10 destinations	Replace a destination or reduce scope transparently	Phase 2
R2	Inconsistent or irrelevant reviews	High / Medium / High	Inspect samples; document relevance and exclusion rules	Flag or exclude unusable records and report counts	Phase 2
R3	Poor sentiment or aspect classification	Medium / High / High	Pilot on a manageable sample; manually validate outputs	Use a simpler baseline or reduce the NLP scope	Phase 3
R4	Language-related model errors	High / Medium / High	Check model language support; review results per language	Restrict advanced analysis to validated languages	Phase 3
R5	Limited time and initial skill level	High / High / High	Include learning time; review progress weekly; start integration early	Drop optional features while protecting the core Power BI deliverable	All phases
R6	Incomplete dates, ratings, or inconsistent source fields	High / Medium / High	Create a field-availability matrix and data-quality checks	Omit unsupported trends or rating statistics	Phase 2
R7	Sampling bias and platform differences	High / Medium / High	Report platform and sample sizes; avoid generalizing to all tourists	State limitations beside comparative metrics	Phases 1 and 4

ID	Risk	Probability / Impact / Priority	Preventive measures	Contingency / fallback	Responsible phase
R8	Misinterpretation of raw complaint counts	Medium / High / High	Use normalized rates with explicit denominators	Add metric definitions and explanatory tooltips	Phase 4
R9	Third-party tool, Actor, price, or free-credit changes	Medium / Medium / Medium	Keep exports and collection logs; check current documentation	Use collected data and simplify the sample if free credits expire	Phase 2
R10	Same review collected via two tools or repeated exports	Medium / Medium / Medium	Match platform review IDs or canonical review URLs where available	Flag records that cannot be resolved and document the uncertainty	Phase 2

1.12 Key Performance Indicators

All targets are *proposed*. No measured value exists yet.

KPI	Proposed target	Measurement method	Actual result	Status
Destinations represented	About 5	Count of destinations with usable data	To be measured	Planned
Usable reviews per destination	About 500-1,000 where feasible	Count after cleaning	Pilot exports: 500 Tripadvisor + 220 Google Maps reviews (raw); cleaned usable counts pending	In progress
Records successfully processed	Provisional: at least 95%	Processed / imported rows	To be measured	Planned
Completeness of key fields	To be set after schema inspection	Share of non-missing values	To be measured	Planned
Technical duplicates detected	Reported, no target	Flagged count / total	To be measured	Planned
Records categorized (NLP)	Provisional target set after pilot	Categorized / eligible	To be evaluated	Planned
Macro F1 on labeled sample	Provisional, set after baseline	scikit-learn on manual labels	To be evaluated	Planned
Aspect precision and recall	Reported where measurable	Manual comparison	To be evaluated	Planned
Dashboard load/interaction time	Measured on project machine	Observed timing	To be measured	Planned
Required pages and filters complete	100% of defined scope	Checklist	To be measured	Planned
Source documentation completeness	All sources documented	Source inventory review	To be measured	Planned

Response time on the project computer is relevant to a desktop prototype. Uptime and user adoption are not applicable to an undeployed prototype and are listed only as future operational KPIs.

1.13 Project Assumptions and Dependencies

- The six-week window is approximate; the institutional deadline will be confirmed separately.
- Outscraper free credits or quotas may limit collection; actual capabilities will be verified in current documentation.
- Not every source supplies every field.
- Pretrained models require free compute; if impractical, the fallback applies.
- Requirements are based on the project brief, not on Ministry interviews.
- All team members can install Power BI Desktop on their computers.

2. Literature Review

This focused review distinguishes published research and official technical documentation from the team's own project assumptions.

2.1 Introduction

Online reviews give visitors a way to describe experiences in their own words, and sentiment analysis research has long treated such text as a source of opinion data (Pang & Lee, 2008; Liu, 2012). A numerical rating summarizes satisfaction in one number; review text can explain what a visitor liked or disliked. Xiang et al. (2015) analyzed a large set of hotel reviews with text analytics and showed that textual content and satisfaction ratings can be examined together, and that text exposes experience dimensions a rating alone does not.

Review-based analysis also has limits. Reviewers are self-selected, platforms differ, and ratings distributions are often uneven. Hu et al. (2009) discuss how product-review ratings tend to follow a J-shaped distribution, with many extreme ratings, which makes simple averages a weak summary. For TourSights this means review data indicates patterns among reviewers, not the opinions of all tourists.

2.2 Online Reviews in Tourism Analytics

Research on tourist and hotel reviews uses text mining to identify recurring themes and link them to satisfaction (Xiang et al., 2015). The relevance to TourSights is the idea that frequent topics in review text can be counted and compared. As volume grows, manual reading becomes impractical because a reader cannot consistently tally themes across platforms, languages, and time periods.

A limitation is that most cited work concerns hotels or products rather than archaeological sites and museums, whose experience factors (queues, guides, vendor behavior, exhibition organization) differ. The proposed taxonomy is therefore the team's own adaptation and must be refined against the actual data rather than treated as an established standard.

2.3 Sentiment Analysis

Sentiment analysis estimates the opinion polarity expressed in text. Pang and Lee (2008) and Liu (2012) describe the main tasks, from document-level polarity to finer-grained opinion mining, and note the difficulty of context, negation, and implicit opinion. Typical labels are positive, negative, and neutral; some applications add a mixed class for reviews that contain both praise and criticism.

A sentiment label is a statistical estimate. It can be wrong for sarcasm, negation, or domain-specific wording. TourSights will therefore validate any classifier on a manually labeled sample and report errors, rather than assuming published performance transfers.

2.4 Aspect-Based Sentiment Analysis

Aspect-based sentiment analysis (ABSA) identifies the aspects of an entity that a text mentions and the opinion expressed toward each. Hu and Liu (2004) were early in summarizing customer reviews by product features. The SemEval-2016 ABSA task (Pontiki et al., 2016) formalized sub-tasks such as aspect category detection and sentiment per aspect, and a recent survey (Zhang et al., 2023) organizes ABSA tasks and methods.

A single review may say that a site is magnificent (positive about the historical experience) while the queue was long and prices were high (negative about queues and price). A review-level label would hide this. Aspect-level analysis separates these signals and shows which aspects are associated with dissatisfaction, which is why TourSights treats it as its analytical goal.

The project's initial approach is deliberately simpler than research ABSA models: predefined categories, transparent keyword or phrase rules, and a pretrained sentiment model, with manual review. This is a pragmatic approximation. Keyword matching can miss context, and a keyword match does not prove a complaint exists.

2.5 Multilingual NLP

Pretrained transformer encoders such as BERT (Devlin et al., 2019) and their multilingual successors such as XLM-R (Conneau et al., 2020) allow a single model to process many languages, and XLM-T (Barbieri et al., 2022) adapts a multilingual model for sentiment on social-media text in several languages. These lines of work suggest that pretrained models can be applied without training from scratch, using libraries such as Hugging Face Transformers (Wolf et al., 2020).

Challenges remain. Arabic has substantial dialect variation and morphological complexity (Darwish et al., 2021), reviews mix scripts and languages, sarcasm and negation differ across languages, and translation can lose sentiment cues. Model quality also varies by language and domain. The project hopes to cover English, Arabic, French, and German, but will report supported and unsupported languages after checking model documentation and validating on its own labeled sample. Equal data volume across languages is not assumed.

2.6 Business Intelligence and Power BI

Microsoft documentation describes Power Query as a data-connection and transformation tool used for extracting, shaping, and loading data (Microsoft, n.d.-a), which supports its role in TourSights as the primary table-cleaning step. Microsoft's guidance on star schemas recommends separating fact tables from dimension tables to give a model that is easier to understand and to filter (Microsoft, n.d.-b), which motivates FactReviews, FactReviewAspects, and dimension tables. DAX is the formula language for measures (Microsoft, n.d.-c), and drill-through pages let users move from a summary to detail for a selected item (Microsoft, n.d.-d).

For decision support, these features mean a dashboard can show consistent measures with explicit denominators, filter by destination, source, and period, and let users inspect underlying records. This is a design rationale drawn from vendor documentation, not evidence that the final dashboard will be effective; usability will be tested during Week 6.

2.7 Existing Tools and Approaches

Commercial review-management and social-listening platforms and general BI dashboards already display ratings and sentiment for businesses. TourSights does not claim to be the first tourism analytics system. What is useful from existing approaches: aggregating reviews, showing sentiment by theme, and filtering by time. What the project adds is a small, transparent, destination-focused prototype built with free tools and aimed at a specific institutional stakeholder.

For data collection, Outscraper documents endpoints for Google Maps reviews, Tripadvisor reviews, YouTube comments, Reddit comments, Booking.com reviews, Expedia reviews, Yelp reviews, and other review/comment sources (Outscraper, n.d.-a). Outscraper also documents a Tripadvisor Reviews API with fields including rating, title, text, author, and date (Outscraper, n.d.-b). Apify provides individual Store Actors, including a Google Maps Reviews Scraper maintained by Compass and a Tripadvisor Reviews Scraper maintained by Max (Apify, n.d.-a, n.d.-b). Their published pages describe review-text, rating, date, and source URL fields, with additional fields depending on the Actor and input. These capabilities are documented, but only the two Outscraper pilot exports are confirmed as collected; the Apify Actors remain planned for a small feasibility test. The exact output schema and obtainable volume must be checked in an actual run.

2.8 Research Gap and Project Contribution

The intended contribution of TourSights is an integrated workflow, for selected Egyptian museums and archaeological sites, that combines review collection, structured preparation, sentiment and aspect categorization where feasible, destination comparison with explicit denominators, and a focused Power BI interface. This is the project's intended contribution, not a claim that no similar product exists. It differs from a dashboard that only shows average ratings because it aims to show which aspects are mentioned, how often they are associated with negative sentiment, and how reliable the underlying sample is.

2.9 Feedback and Evaluation: Lecturer's Assessment

No formal lecturer assessment had been received at the time of preparing this report. Formal feedback and any resulting changes will be incorporated after the project supervisor's review.

2.10 Suggested Improvements

The following are the team's own proposed future improvements. They are not completed features and are not lecturer suggestions.

- Wider destination coverage and additional verified data sources.
- Better multilingual classification and a larger manually labeled evaluation set.
- Improved aspect-level accuracy.
- Automated, scheduled refresh.
- More advanced trend and change detection.
- Validation with tourism experts and testing with intended institutional users.
- Improved dashboard navigation.

- Integration of verified operational data where available.

2.11 Final Grading Criteria

The table below reproduces the grading framework supplied by the initiative. It is a general rubric; the relevance and detailed application of individual criteria may depend on the project's nature and the supervisor's instructions. No marks have been awarded.

Criterion	Weight
Idea & Objectives	10/100
System Analysis & Requirements / Design	15/100
Algorithms / Methods / Techniques	20/100
Implementation & Coding / Results	20/100
Testing & Documentation	10/100
Innovation & Added Value	10/100
Presentation & Demo	10/100
Final Submission & Delivery	5/100
TOTAL	**100/100**

Planned evidence for each criterion.

Criterion	Planned evidence from TourSights
Idea & Objectives	Sections 1.2-1.3; scope and stakeholder definition
System Analysis & Requirements / Design	Section 3; data model; architecture
Algorithms / Methods / Techniques	Duplicate rule, taxonomy, sentiment and aspect method, validation procedure
Implementation & Coding / Results	Power Query workflow, NLP notebook, .pbix file; results only after they are measured
Testing & Documentation	Section 5 test table; data dictionary; this report
Innovation & Added Value	Aspect-level, normalized, multi-destination view for a ministry-level audience
Presentation & Demo	Data-to-dashboard demonstration in Week 6
Final Submission & Delivery	Final package listed in Phase 4 handover

3. Requirements Gathering

3.1 Requirements Gathering Method

Requirements are derived from: the academic project brief; the project objectives; the intended Ministry-level stakeholder; the data-availability assessment (in progress); the team's implementation constraints; the desired analytical outputs; and the six-week schedule. No interviews with Ministry representatives have taken place, and direct stakeholder access has not been confirmed. The requirements therefore reflect the brief and the intended use and should be refined through stakeholder feedback if access becomes available.

3.2 Stakeholder Analysis

Stakeholder	Type	Interest and information needs	Expected interaction
Decision-makers, Ministry of Tourism and Antiquities	Primary (intended)	Frequent issues, comparison across destinations, trends, positives, evidence behind figures	Views overview and destination pages; uses filters and drill-through
Destination managers	Secondary	Aspect-level issues for their own site	Views a destination page

Stakeholder	Type	Interest and information needs	Expected interaction
Tourism and performance analysts	Secondary	Metric definitions, data coverage, limitations	Examines measures and data documentation
Project team	Internal	Achievable scope; correct, tested outputs	Builds, tests, documents
Instructor and evaluators	Academic	Rigor, honesty about results, documentation	Reviews report, demo, and files

No stakeholder has formally approved these requirements.

3.3 User Needs

The primary user needs to: understand the main negative aspects of visitor experience; compare selected destinations; monitor trends where dates are reliable; identify frequently mentioned issues; see positive aspects as well as complaints; filter by destination and time period; access relevant underlying review details when available; and understand dataset size, source coverage, and analytical limitations.

3.4 User Stories

ID	User story	Scope
US-01	As a Ministry decision-maker, I want an overview of visitor feedback across destinations, so that I can see where attention may be needed.	In scope
US-02	As a decision-maker, I want a page for each destination, so that I can examine one site's feedback in detail.	In scope
US-03	As a decision-maker, I want to compare destinations using normalized measures, so that larger samples do not make a site look worse by count alone.	In scope
US-04	As an analyst, I want to see the most common complaint categories, so that I can find recurring issues.	In scope (if NLP implemented)
US-05	As a decision-maker, I want to see complaint patterns over time, so that I can notice changes.	In scope where dates suffice
US-06	As a manager, I want to see positive experiences, so that strengths are not overlooked.	In scope
US-07	As a user, I want to filter by destination, period, source, and language, so that I can focus on relevant data.	In scope where fields exist
US-08	As an analyst, I want to drill through to underlying reviews, so that I can check the evidence.	In scope where feasible
US-09	As any user, I want to see sample sizes, sources, and limitations, so that I interpret results correctly.	In scope
US-10	As a decision-maker, I want automatic scheduled updates, so that data stays current.	Future work

3.5 Use Cases

ID	Name	Actor	Preconditions	Main flow	Expected outcome	Limitations
UC-01	View overview	Decision-maker	Report open	Open Overview page	See coverage, sentiment, top issues, comparison	Depends on available data
UC-02	Select a destination	Decision-maker	Report open	Click destination navigation	Destination page shown	One page per finalized destination
UC-03	Filter by period or source	Analyst	Fields available	Use slicers	Visuals update	Only for supported fields
UC-04	Compare destinations	Decision-maker	Normalized measures built	Open comparison visuals	See rates with denominators	Sample sizes differ

ID	Name	Actor	Preconditions	Main flow	Expected outcome	Limitations
UC-05	Investigate an aspect	Analyst	NLP output present	Select an aspect	See mentions, negative rate, trend	Keyword/model errors possible
UC-06	Review drill-through evidence	Analyst	Drill-through page implemented	Right-click a visual, drill through	See selected review details	Only if source permits and data supports
UC-07	Understand method and sample	Any user	Report open	Open information/notes	See sources, date range, definitions, limits	Reflects actual dataset only

3.6 Functional Requirements

Priority: M = Must Have, S = Should Have, C = Could Have. Objective references are to Section 1.3.

ID	Description	Priority	Rationale	Acceptance criterion	Objective
FR-01	Ingest review data from the tested collection source	M	Data foundation	Source files import without error	O2
FR-02	Store destination, text, date, rating, source, and language where available	M	Analysis needs these fields	Fields present or documented as unavailable	O2, O3
FR-03	Perform structured cleaning in Power Query	M	Reproducible preparation	Documented query steps run on refresh	O3
FR-04	Distinguish technical duplicates from independent reviews	M	Preserve complaint frequency	Rule documented; flagged records listed; similar-text reviews retained	O4
FR-05	Provide consistent destination and source categories	M	Valid comparison	Zero unmapped values after cleaning	O3
FR-06	Calculate aggregate review statistics	M	Core indicators	Totals reconcile with source table	O6
FR-07	Provide sentiment and aspect classifications if NLP is implemented	S	Analytical extension	Output file joined by review ID; metrics measured on labeled sample	O5
FR-08	Report actual usable reviews per destination	M	Context for interpretation	Counts shown on every destination page	O2, O6
FR-09	Calculate negative measures with explicit denominators	M	Avoid misinterpretation	Each measure defined in documentation and tooltip	O6
FR-10	Display most frequently mentioned issue categories	S	Key output	Visual matches category counts in data	O5, O6
FR-11	Display positive visitor-experience aspects	S	Balanced view	Positive-aspect visual present	O5, O6
FR-12	Compare destinations with normalized measures	M	Fair comparison	Rates shown with denominators	O6
FR-13	Provide trend visuals when dates are sufficient	S	Monitor change	Shown only above a documented data-sufficiency rule	O6
FR-14	Allow filtering by destination and other supported fields	M	Usability	Slicers work and totals respond	O6
FR-15	Provide one principal page per final destination	M	Project design	Page exists for each finalized destination	O6
FR-16	Provide drill-through detail pages where appropriate	C	Evidence access	Navigation works for tested cases	O6
FR-17	Document dataset, sources, method, validation, and limitations	M	Transparency	Sections of the report complete	O7
FR-18	Distinguish measured results from targets and future work	M	Honesty	Labels applied throughout	O7

ID	Description	Priority	Rationale	Acceptance criterion	Objective
FR-19	Provide a priority indicator with a documented rule	C	Prioritization	Rule, weights, and limits documented; not described as an official risk rating	O6

Automated refresh, alerts, and a public web application are not requirements for this version.

3.7 Non-Functional Requirements

ID	Category	Requirement	Acceptance method	Priority
NFR-01	Usability	A non-technical decision-maker can follow the main pages	Walkthrough with at least one non-team volunteer	S
NFR-02	Performance	Loading and interaction time measured on the project computer	Recorded observation; no target assumed	S
NFR-03	Reliability	Same eligible data produces the same outputs	Repeat refresh and compare totals	M
NFR-04	Data quality	Key fields, missing values, duplicates, and source coverage documented	Quality report	M
NFR-05	Maintainability	Clear field names, documented measures, reproducible transformation steps	Peer review	S
NFR-06	Privacy	No unnecessary personal data; reviewer identifiers not displayed	Field review	M
NFR-07	Transparency	Source, date range, sample size, definitions, limits are visible or documented	Checklist	M
NFR-08	Resource efficiency	Free tools; runs on ordinary student hardware	Setup check	M
NFR-09	Extensibility	New destinations or sources can be added without redesigning the model	Design review	C
NFR-10	Interpretation	NLP outputs are not presented as guaranteed facts	Wording review	M

3.8 Data Requirements and Data Dictionary

Not every source provides every field. Status: R = required if available, O = optional, D = derived during processing.

Field	Description	Type	Status	Quality check
ReviewID	Stable source review identifier; where unavailable, a defensible identifier may be derived from the review URL	Text	R if available / D otherwise	Unique after technical duplicate checks
DestinationID	Internal destination code	Text	D	Maps to DimDestination
DestinationName	Standardized tourist-site or museum name	Text	R	No unmapped destination values
Platform	Platform where the review was originally published (e.g., Google Maps or Tripadvisor)	Text	R	Matches DimSource platform values
CollectionTool	Tool used to obtain the record (e.g., Outscraper or Apify)	Text	D	Matches the collection log
ReviewText	Review content in the available source language	Text	R	Not empty for included text analysis
ReviewTitle	Review headline, if supplied by the source	Text	O	Optional; do not infer if absent

Field	Description	Type	Status	Quality check
PublicationDate	Date the review was published	Date/time	R if available	Valid date; missingness reported
CollectionDate	Date the export was obtained	Date	D/O	Recorded from collection log
Rating	Numerical star rating supplied by the platform	Number	O	Within source rating scale
Language	Source-provided or automatically detected language	Text	O/D	Supported language labels documented
ReviewURL	URL of the original review, if included	Text	O	URL format and duplication check
TechnicalDuplicateFlag	Flag for the same review repeated through extraction or merging	Boolean	D	Rule documented; similar independent reviews retained
OverallSentiment	Positive, Neutral, Negative; Mixed only where method supports it	Text	D	Allowed labels; measured validation
AspectCategory	Visitor-experience category detected by model/rules	Text	D	Must belong to documented taxonomy
AspectSentiment	Sentiment toward a given aspect, if supported	Text	D	Allowed labels; manual validation
Confidence	Model score, if produced	Number	O/D	Not assumed to be a calibrated probability

The actual columns of the Giza pilot file will be inspected in Phase 2 and the dictionary will be updated to match.

3.9 Proposed Data Model

Table	Type	Grain / key	Purpose
FactReviews	Fact	One row per review (ReviewID)	Review-level attributes and overall sentiment
FactReviewAspects	Fact	One row per review-aspect pair	Aspect mentions and aspect sentiment
DimDestination	Dimension	DestinationID	Destination attributes
DimSource	Dimension	SourceID	Platform
DimDate	Dimension	Date	Time analysis
DimAspect	Dimension	AspectID	Taxonomy
DimLanguage	Dimension	LanguageID	Language

One review can map to several rows in FactReviewAspects. Review counts must therefore be computed from FactReviews (or with distinct counts), while aspect-mention counts come from FactReviewAspects. Relationships run from dimensions to facts. The model will be adapted to the fields actually available.

Metric definitions (proposed).

Metric	Numerator	Denominator
Negative review rate	Reviews classified Negative	All usable reviews in the filter
Aspect negative rate	Mentions of aspect A with negative sentiment	All mentions of aspect A
Reviews mentioning aspect	Distinct reviews with aspect A	(count, or share of all usable reviews)

Metric	Numerator	Denominator
Aspect mention count	Rows in FactReviewAspects for A	(count)

These are distinct quantities and must not be conflated. Representative DAX formulas, if included in the final report, will be labeled as proposed examples until tested.

3.10 Acceptance Criteria and Traceability

Objective	Stakeholder need	Requirements	Dashboard feature	Planned test	KPI
O1	Clear scope	FR-17, FR-18	Notes page	T-13	Documentation completeness
O2	Data availability	FR-01, FR-02, FR-08	Sample-size cards	T-01, T-02	Usable reviews per destination
O3	Reliable data	FR-03, FR-05	Data-quality notes	T-03, T-05, T-06	Field completeness
O4	Faithful frequencies	FR-04	Flag in model	T-04	Duplicate rate
O5	Issue identification	FR-07, FR-10, FR-11	Aspect visuals	T-07, T-08	Macro F1; precision/recall
O6	Comparison and trends	FR-06, FR-09, FR-12 to FR-16, FR-19	Overview, destination, drill-through	T-09 to T-12	Pages and filters complete
O7	Trust and evidence	FR-17, FR-18, NFR-07	Notes and limitations	T-13	Documentation completeness

4. Proposed System Architecture and Analytical Workflow

Layer	Component	Role	Status
Acquisition	Outscraper + Apify Actors where tested	Export reviews from selected platforms to CSV/Excel; store platform and collection-tool separately	Outscraper pilot exports: 500 Tripadvisor + 220 Google Maps reviews; Apify planned, not run
Storage	Raw folder and collection log	Untouched source files	Planned
Preparation	Power Query	Standardize names, types, destinations, sources, dates, ratings; flag technical duplicates; document steps	Planned
Enrichment (optional)	Python, pretrained model, keyword rules	Sentiment and aspect labels, joined by ReviewID	Planned; depends on validation
Modeling	Power BI star schema	FactReviews, FactReviewAspects, dimensions	Planned
Presentation	DAX measures and report pages	Overview, destination pages, hidden drill-through	Planned
Quality	Manual sample and test table	Validation of NLP outputs and dashboard totals	Planned

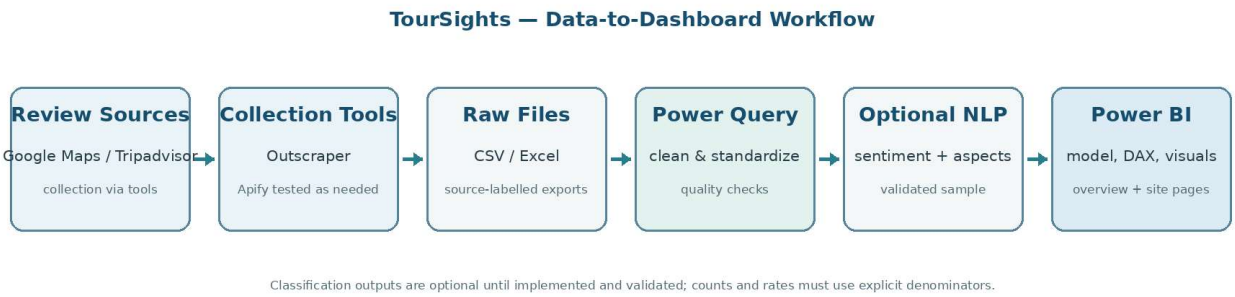


Figure 1. Proposed TourSights workflow; Apify collection and NLP enrichment remain planned until tested.

Analytical logic. (1) Overall sentiment per review. (2) Detection of aspect categories using a documented taxonomy and keyword rules, with manual review of ambiguous cases. (3) Aspect-level sentiment where the method supports it. (4) Aggregation into counts and rates with stated denominators. (5) Destination comparison using rates rather than raw counts. (6) An optional priority indicator combining frequency, negative rate, change over time, and data volume under a documented rule; it is an analytical aid and not an official risk rating.

Model selection. A multilingual pretrained sentiment model such as cardiffnlp/twitter-xlm-roberta-base-sentiment (based on XLM-R and fine-tuned for sentiment on multilingual social-media text) is a candidate to evaluate. Its model card reports fine-tuning on eight languages, including Arabic, English, French, and German (Cardiff NLP, n.d.). Because its training domain is Twitter rather than tourist reviews, suitability must be assessed on a manually labeled sample of the project data; the report will not assume that benchmark results transfer to this dataset. License and runtime requirements must be checked before use.

Fallback. If multilingual quality is unacceptable, restrict classification to validated languages, or use a simpler documented baseline (for example, rating-based polarity and keyword categories) so the core dashboard remains deliverable.

Dashboard structure (proposed). Page 1: overview and destination comparison. Pages 2 onward: one visible page per finalized destination. Hidden pages: drill-through review detail where permitted by the source and supported by the data. Hidden pages are a navigation choice, not data protection.

4.1 Data Collection Tool Configuration

The report separates the platform on which a review was published from the tool used to collect it. The two Outscraper exports are the only confirmed pilot collection results. Apify is an additional planned option; its relevant Actors will be tested on a small sample before their outputs are merged into the project dataset. Actor pages document possible fields and export formats, but do not establish that a particular destination will return a sufficient or clean sample (Outscraper, n.d.-a, n.d.-b; Apify, n.d.-a, n.d.-b).

Review platform	Collection tool	Status at 19 October 2026	Potential fields and role
Tripadvisor	Outscraper	Confirmed: 500 reviews for The Pyramids of Giza in pilot export	Review text/title, rating, date, review URL where supplied; raw count not yet cleaned
Google Maps	Outscraper	Confirmed: 220 reviews for Giza Necropolis in pilot export	Review text, rating, publication date and review URL where supplied
Tripadvisor	Apify: maxcopell/tripadvisor-reviews	Planned small test; no run completed	Actor page documents review text, rating, publication date, travel date, language and URL; only load fields needed for analysis
Google Maps	Apify: compass/google-maps-reviews-scraper	Optional fallback/test; no run completed	Could supplement or validate Google Maps collection; check overlap before combining

5. Testing and Validation Plan

Actual results and pass/fail status are left unfilled until tests are run.

Test ID	Objective	Input / condition	Expected result	Actual result	Pass/Fail
T-01	Data import	Raw CSV/Excel files	All files load; row counts match source	To be completed during testing	To be completed during testing
T-02	Missing values	Cleaned table	Missing counts per field documented	To be completed during testing	To be completed during testing
T-03	Destination mapping	Destination column	No unmapped destination names	To be completed during testing	To be completed during testing
T-04	Duplicate records	Repeated IDs; similar-text reviews	Only technical duplicates flagged; similar-text reviews retained	To be completed during testing	To be completed during testing
T-05	Date and rating consistency	Date and rating columns	Valid dates; ratings in scale	To be completed during testing	To be completed during testing
T-06	Totals cross-check	Dashboard totals vs source tables	Totals reconcile	To be completed during testing	To be completed during testing
T-07	Sentiment validation	Manually labeled sample	Metrics computed and reported	To be completed during testing	To be completed during testing
T-08	Aspect validation	Manually labeled sample	Precision/recall computed per category	To be completed during testing	To be completed during testing
T-09	DAX verification	Measures vs manual calculation	Values agree; no double counting	To be completed during testing	To be completed during testing
T-10	Filters and slicers	Combinations of selections, empty selections	Visuals respond correctly	To be completed during testing	To be completed during testing
T-11	Destination pages	Each page	Correct destination data	To be completed during testing	To be completed during testing
T-12	Drill-through	Selected visuals	Correct filtered details	To be completed during testing	To be completed during testing
T-13	Usability, loading, documentation	Walkthrough; timing on project computer; checklist	Understandable pages; times recorded; documentation complete	To be completed during testing	To be completed during testing

NLP validation procedure. Select a manageable sample of actual reviews; write annotation guidelines; have members label independently where feasible; resolve disagreements by documented rules; keep development examples separate from the final evaluation sample where practical; compare model output with the labels; review common errors; report limitations. Metrics (precision, recall, F1, macro F1, confusion matrix) are *proposed evaluation metrics* until measured. Any numerical acceptance target will be a provisional project target set after a baseline run.

6. Limitations and Assumptions

- Online reviewers are not necessarily representative of all tourists, and a large sample does not by itself remove sampling bias.
- Coverage differs by destination and platform; platform populations and rating habits differ.
- Reviews may be outdated, influenced by individual circumstances, or may not describe a direct visit.
- Language and translation issues may reduce accuracy; sentiment models can mishandle sarcasm, negation, and mixed opinions.
- Keyword rules can misclassify context; categories can overlap.
- Review text often cannot identify the exact physical location of a problem.
- A higher negative review count does not mean a destination is objectively worse.
- Correlation in reviews does not prove causation.
- The priority indicator is an analytical aid, not an official risk assessment.
- No consultation with Ministry personnel has taken place, which limits operational validation.

- The system supports identification of patterns for further evaluation by responsible officials; it does not replace field inspections, verified operational statistics, or professional expertise.

Assumptions. See Section 1.13.

7. Future Work

The items below are future work, not current capabilities. The current prototype is a static analytical snapshot with an interactive Power BI report.

- Scheduled data refresh and automated repeat collection.
- More destinations and additional verified sources.
- More robust multilingual aspect-based models and a larger labeled evaluation set.
- Detection of emerging issues and alerts for meaningful changes in complaint rates; historical comparisons.
- Integration with verified operational information.
- Validation with tourism specialists and usability testing with intended institutional users.
- A production deployment, if approved and justified.

8. Conclusion

TourSights proposes a practical workflow for turning online visitor reviews of selected Egyptian museums and archaeological sites into structured indicators within an interactive Power BI report. The planned workflow combines collection through Outscraper and an evaluation of Apify, Power Query cleaning, optional sentiment and aspect analysis, and destination comparison using explicit denominators. The implementation strategy is scaled to approximately six weeks and a four-member team with limited NLP experience. At the time of this report, two Outscraper pilot exports are available: 500 Tripadvisor reviews for The Pyramids of Giza and 220 Google Maps reviews for Giza Necropolis. These are extraction counts, not yet cleaned unique-record counts. No measured model accuracy, completed dashboard result, or improvement in tourist satisfaction has been demonstrated. The project is intended to identify patterns for further evaluation; decisions remain with the responsible institution.

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Appendices

Appendix A - Preliminary Data Dictionary

This appendix consolidates the proposed fields for the unified analytical schema. “Platform” identifies where a review was published; “CollectionTool” identifies how it was collected. Actual source columns must be mapped only when present in the export. R = required if available, O = optional, D = derived.

Field	Description	Type	Status	Quality check
ReviewID	Stable source review identifier; where unavailable, a defensible identifier may be derived from the review URL	Text	R if available / D otherwise	Unique after technical duplicate checks
DestinationID	Internal destination code	Text	D	Maps to DimDestination
DestinationName	Standardized tourist-site or museum name	Text	R	No unmapped destination values
Platform	Platform where the review was originally published (e.g., Google Maps or Tripadvisor)	Text	R	Matches DimSource platform values
CollectionTool	Tool used to obtain the record (e.g., Outscraper or Apify)	Text	D	Matches the collection log
ReviewText	Review content in the available source language	Text	R	Not empty for included text analysis
ReviewTitle	Review headline, if supplied by the source	Text	O	Optional; do not infer if absent
PublicationDate	Date the review was published	Date/time	R if available	Valid date; missingness reported
CollectionDate	Date the export was obtained	Date	D/O	Recorded from collection log
Rating	Numerical star rating supplied by the platform	Number	O	Within source rating scale
Language	Source-provided or automatically detected language	Text	O/D	Supported language labels documented
ReviewURL	URL of the original review, if included	Text	O	URL format and duplication check
TechnicalDuplicateFlag	Flag for the same review repeated through extraction or merging	Boolean	D	Rule documented; similar independent reviews retained
OverallSentiment	Positive, Neutral, Negative; Mixed only where method supports it	Text	D	Allowed labels; measured validation
AspectCategory	Visitor-experience category detected by model/rules	Text	D	Must belong to documented taxonomy
AspectSentiment	Sentiment toward a given aspect, if supported	Text	D	Allowed labels; manual validation
Confidence	Model score, if produced	Number	O/D	Not assumed to be a calibrated probability

Table A1. Preliminary unified data dictionary; subject to the actual exported fields.

Appendix B - Requirements Traceability Matrix

The matrix below connects project objectives to stakeholder needs, functional requirements, dashboard features, planned tests, and success indicators.

Objective	Stakeholder need	Requirements	Dashboard feature	Planned test	KPI
O1	Clear scope	FR-17, FR-18	Notes page	T-13	Documentation completeness
O2	Data availability	FR-01, FR-02, FR-08	Sample-size cards	T-01, T-02	Usable reviews per destination
O3	Reliable data	FR-03, FR-05	Data-quality notes	T-03, T-05, T-06	Field completeness
O4	Faithful frequencies	FR-04	Flag in model	T-04	Duplicate rate
O5	Issue identification	FR-07, FR-10, FR-11	Aspect visuals	T-07, T-08	Macro F1; precision/recall
O6	Comparison and trends	FR-06, FR-09, FR-12 to FR-16, FR-19	Overview, destination, drill-through	T-09 to T-12	Pages and filters complete
O7	Trust and evidence	FR-17, FR-18, NFR-07	Notes and limitations	T-13	Documentation completeness

Table B1. Requirements traceability matrix.

Appendix C - Risk Register

The risk register below is the working register for the current prototype. Likelihood, impact, and priority are qualitative planning judgments, not statistical estimates.

ID	Risk	Probability / Impact / Priority	Preventive measures	Contingency / fallback	Responsible phase
R1	Insufficient data for some destinations	Medium / High / High	Early feasibility tests; maintain a candidate pool of 8-10 destinations	Replace a destination or reduce scope transparently	Phase 2
R2	Inconsistent or irrelevant reviews	High / Medium / High	Inspect samples; document relevance and exclusion rules	Flag or exclude unusable records and report counts	Phase 2
R3	Poor sentiment or aspect classification	Medium / High / High	Pilot on a manageable sample; manually validate outputs	Use a simpler baseline or reduce the NLP scope	Phase 3
R4	Language-related model errors	High / Medium / High	Check model language support; review results per language	Restrict advanced analysis to validated languages	Phase 3
R5	Limited time and initial skill level	High / High / High	Include learning time; review progress weekly; start integration early	Drop optional features while protecting the core Power BI deliverable	All phases
R6	Incomplete dates, ratings, or inconsistent source fields	High / Medium / High	Create a field-availability matrix and data-quality checks	Omit unsupported trends or rating statistics	Phase 2
R7	Sampling bias and platform differences	High / Medium / High	Report platform and sample sizes; avoid generalizing to all tourists	State limitations beside comparative metrics	Phases 1 and 4
R8	Misinterpretation of raw complaint counts	Medium / High / High	Use normalized rates with explicit denominators	Add metric definitions and explanatory tooltips	Phase 4

ID	Risk	Probability / Impact / Priority	Preventive measures	Contingency / fallback	Responsible phase
R9	Third-party tool, Actor, price, or free-credit changes	Medium / Medium / Medium	Keep exports and collection logs; check current documentation	Use collected data and simplify the sample if free credits expire	Phase 2
R10	Same review collected via two tools or repeated exports	Medium / Medium / Medium	Match platform review IDs or canonical review URLs where available	Flag records that cannot be resolved and document the uncertainty	Phase 2

Table C1. Risk register and mitigation actions.

Appendix D - Project Timeline and Deliverables

The dates are a proposed six-week schedule beginning 19 October 2026, not a confirmed institutional deadline. The team should adjust this schedule if the official submission date differs.

Week	Dates	Main activities	Deliverable
1	19-25 Oct 2026	Confirm problem, scope, initial destinations; test sources and fields; define data schema; confirm initial categories	Approved project plan; data-feasibility results
2	26 Oct-1 Nov 2026	Collect reviews for pilot destinations; expand where data allows; document sources and limits	Initial destination datasets; source inventory
3	2-8 Nov 2026	Clean and standardize in Power Query; validate dates, ratings, languages, identifiers; resolve technical duplicates; refine categories	Cleaned analytical dataset
4	9-15 Nov 2026	Evaluate pretrained models; implement feasible sentiment/aspect approach; label validation sample; evaluate	Initial model outputs and evaluation (if successful)
5	16-22 Nov 2026	Build Power BI model and DAX measures; overview and destination pages; drill-through where feasible	Working dashboard prototype
6	23-29 Nov 2026	Validate dashboard and results; fix major issues; finish report and presentation; prepare demo	Validated prototype and final documentation

Table D1. Proposed six-week execution timeline.

Milestone	Planned period	Main activities	Deliverable	Completion criteria
M1 Plan and scope approved	Week 1	Scope, objectives, candidate pool, source tests	Project plan; source matrix	Scope and exclusions documented; owners assigned
M2 Raw data collected	Weeks 1-2	Outscraper collection; Apify Actors for small tests; collection log	Raw files per destination	Actual counts recorded per destination
M3 Cleaned dataset	Weeks 2-3	Power Query cleaning; quality checks	Cleaned table; quality report	Types standardized; duplicate rule documented
M4 NLP outputs (if feasible)	Weeks 3-4	Model choice; categories; manual labeling	Enriched CSV; evaluation notes	Metrics computed from real labeled sample
M5 Dashboard prototype	Weeks 4-5	Model, DAX, pages	.pbix prototype	Overview and destination pages open and work
M6 Validated final package	Week 6	Testing; report; demo	Final report, .pbix, presentation	Test table completed with actual outcomes

Table D2. Milestones, deliverables, and completion criteria.

Appendix E - Team Responsibilities

Assign one primary phase to each member after the team confirms the allocation. Student IDs and names are intentionally blank for manual completion.

Project Phase / Primary Responsibility	Student ID	Student Name
Phase 1 - Project Planning, Research, and Requirements		
Phase 2 - Data Collection and Data Preparation		
Phase 3 - NLP, Sentiment Analysis, and Analytical Modeling		
Phase 4 - Power BI Dashboard, Testing, and Integration		

Table E1. Four-phase assignment table. Complete only the Student ID and Student Name cells.

Supporting workstreams and peer-review expectations:

Workstream	Phase owner	Support
Coordination, documentation, research	Phase 1	All (review)
Source assessment and collection	Phase 2 (with Phase 1 input)	Phase 1
Data preparation, Power Query, modeling inputs	Phase 2	Phase 4
NLP exploration, validation, analytical methods	Phase 3	All (annotation)
Dashboard, DAX, testing, presentation	Phase 4	All (testing)

Table E2. Workstream ownership and support.

Appendix F - Evaluation Rubric

The initiative-supplied rubric is reproduced below without changing the weights. It is a generic framework; applicability details may depend on project type and supervisor instructions.

Criterion	Weight
Idea & Objectives	10/100
System Analysis & Requirements / Design	15/100
Algorithms / Methods / Techniques	20/100
Implementation & Coding / Results	20/100
Testing & Documentation	10/100
Innovation & Added Value	10/100
Presentation & Demo	10/100
Final Submission & Delivery	5/100
TOTAL	**100/100**

Table F1. Initiative grading rubric.

Table F2. Proposed evidence from TourSights for each criterion.

Criterion	Planned evidence from TourSights
Idea & Objectives	Sections 1.2-1.3; scope and stakeholder definition
System Analysis & Requirements / Design	Section 3; data model; architecture
Algorithms / Methods / Techniques	Duplicate rule, taxonomy, sentiment and aspect method, validation procedure
Implementation & Coding / Results	Power Query workflow, NLP notebook, .pbix file; results only after they are measured
Testing & Documentation	Section 5 test table; data dictionary; this report
Innovation & Added Value	Aspect-level, normalized, multi-destination view for a ministry-level audience
Presentation & Demo	Data-to-dashboard demonstration in Week 6
Final Submission & Delivery	Final package listed in Phase 4 handover

Appendix G - Illustrative Dashboard Layout (proposed design, not an implemented dashboard)

Page	Proposed contents
Overview and comparison	Cards: total usable reviews, destinations covered; sentiment distribution; top negative aspects; normalized negative rate by destination; trend (if dates suffice); positive aspects; slicers: destination, period, source, language
Destination page (one per destination)	Review count and sample note; sentiment distribution; aspect mentions; negative rate by aspect; positive aspects; monthly trend (if supported); rating summary; priority ranking; drill-through button
Hidden drill-through	Selected review text, date, source, aspect, sentiment (if permitted and supported)
Notes	Sources, date range, metric definitions, limitations

Appendix H - Sample Data Structure (illustrative schema only; contains no real reviews)

ReviewID	DestinationName	DataSource	ReviewText	PublicationDate	Rating	Language
(text)	(text)	(text)	(text)	(date)	(number)	(text)

Appendix I - Data Acquisition Sources and Tools

This matrix separates the review platform from the collection tool and records the project status at the report date. The presence of a documented Actor or endpoint does not mean that it has already been run against a target destination.

Review platform	Collection tool / reference	Project status	Data and use in TourSights
Tripadvisor	Outscraper Tripadvisor Reviews API	Confirmed: 500 extracted reviews for The Pyramids of Giza	Review text, title, rating, date, and review URL where supplied; actual field names are checked in the export
Google Maps	Outscraper Google Maps reviews	Confirmed: 220 extracted reviews for Giza Necropolis	Review text, rating, publication date, and review URL where supplied
Tripadvisor	Apify maxcopell/tripadvisor-reviews	Planned small test; no run completed	Actor page documents text, rating, published date, travel date, language and URL; use minimum fields needed
Google Maps	Apify compass/google-maps-reviews-scraper	Optional test/fallback; no run completed	Consider only if it adds value; check cross-tool overlap and keep the platform distinct from collection tool

Table I1. Data acquisition source/tool matrix, based on project progress and official tool documentation reviewed on 19 October 2026.